



Here's what a rover has to navigate through

turned.) On a rocky, dusty surface with slopes, the possibilities of a rover slipping down a slope—backward or forward—is very real. The rovers adjust for this by calculating distances moved using visual odometer software. What this software does is “remember” previously-identified points on its path and then compare the distance moved by looking at before-and-after images to give much more precise calculations.

The rovers are designed this way to avoid bumping over rocks, falling into canyons and running into walls, so as to protect its fragile interiors, which are already running in extreme conditions. Remember, there's no margin for error on a delicate machine that's on another planet, a hundred million kilometres away from us.

Imaging

The whole point of sending unmanned spacecraft to Mars is to photograph it in order to collect as much visual information about the planet before we send humans there. The results that the rovers have sent back to us are the most exciting images of space exploration we've seen till date—it's about as close as we can get to “experiencing” Mars without going there ourselves.

The cameras on the rovers were perhaps the most high-tech in the world at the time they were made, and they're still being improved. Between the two rovers and the Mars Landers that carried them, NASA has been receiving images from 20 different high-quality, high-resolution cameras at regular intervals. However, it's not the hardware that's impressive, it's the software that manipulates and sends back those images that is of more interest.

Developed specially at NASA's Jet Propulsion Lab (JPL), the imaging software uses ICER wavelet-based compression to reduce image sizes by up to 12 times. Each image that is clicked is divided into 30 chunks and transmitted back to Earth—this is to ensure that if packets are dropped, the entire image is not lost. It compresses images progressively, and features Image Context Error Correction, which accounts for any errors that may occur when data is transmitted along the Deep Space Network. Depending on the needs, the software can achieve lossy or lossless compression of images. It's because of this software that we're able to see more pictures that the Mars rovers click, instead of them transmitting fewer RAW images.

Other Technologies

Although there are literally thousands of technologies involved in building spacecraft that are capable of trips to Mars, we'd have to continue this story over the next 400 issues of *Digit* to tell you about them all in detail. Instead, we've clubbed a few interesting technologies together to whet your appetite:

Batteries: As of now, we don't have batteries that can withstand Martian nights, with temperatures of up to minus 110 degrees Celsius. Instead, all batteries on the rovers, and any vehicles to follow, are encased inside insulated boxes with heaters. The batteries can supply power as long as they are above minus 20 degrees Celsius, and need to be hotter than zero degrees Celsius when they are being recharged. Electric heaters and any heat dissipated from other components are used to “warm up” the batteries.

Heaters: The heaters we just mentioned are radioisotope heaters that can produce about a watt of heat for decades. They produce heat through radioactive decay, and using 2.7 grams of plutonium dioxide, encapsulated in plutonium-

rhodium alloy and then locked inside several layers of a carbon-graphite composite.

Space Ship: To go on a round-trip to Mars requires a huge amount of food, water, oxygen and most of all, fuel. It's virtually impossible to send a ship that big into space from Earth. However, building the ship on the Moon, or in orbit in space is possible. Several experts believe that this is the way to go ahead building a ship big enough to go to Mars and back.

Food: This really isn't as big a problem as we think. Food in space flights is just painful to store and eat. Astronauts who already spend weeks in space eat three meals a day, use liquid salt and pepper, and ketchup and mustard to enhance the taste of the food according to their individual tastes. They have to eat out of special containers, of course, to prevent the food from floating away. There's an oven provided to warm up the food, and meals include macaroni and cheese, spaghetti, fruit, brownies, juices, tea, coffee, etc.—seriously, it's not like they're eating little pellets. What's more interesting is that for a really long journey, say, to Mars, small women are more desirable than big, strapping men—a small woman would need about 2,000 calories a day, whereas a macho man would eat over 3,000. Maybe the first humans on Mars will be women, considering that a trip to Mars requires over a year's supply of food!

Air: This is where it all gets complex. Taking the example of the International Space Station (ISS), air is recycled and carbon dioxide removed and “scrubbed” back into oxygen. Oxygen is also produced from electrolysis. In addition, air scrubbers also have to remove ammonia and acetone that the human body produces. Interestingly, scrubbers on the ISS have given trouble before, and reserve oxygen from visiting ships have been used in the past. When travelling to Mars, no such luxuries will be available, so systems to cleanse air and generate oxygen will have to be ~much~ more foolproof and numerous. Another option, of course, is to just use plants to cleanse the air of carbon dioxide and produce oxygen, but again, weight and space constraints will play spoilsport with that idea—not to forget the daily dose of sunlight that plants need in order to survive.

In Summary

A holiday to Mars for the common man is at least a millennium away, but renewed interest in Mars promises to put a man on Mars in a few decades, and most of us will be alive to witness it. Currently, technology is not very far behind what's required to achieve the task, but the prohibitive cost is what's keeping the race to Mars in check. Earlier estimates suggested that it would cost over \$300 billion (12 lakh crore rupees) to get a man to Mars, but now NASA hopes to get the job done using less than \$50 billion (2 lakh crore rupees) after a couple of decades. Considering the costs, there are many who are against the Mission to Mars, citing the good that the same amount of money could do for humanity here on Earth. However, who knows what untold riches Mars holds in store for humanity, in terms of the discovery of life, new minerals, and an invaluable understanding of the history of our solar system? Perhaps even more clues to the eternal questions “Are we alone?” and “Why are we here?”, which humanity certainly cannot place a price tag on? ■

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Image Credit: NASA / JPL / Caltech